NLP

10-06-26

Natural Language Processing :

↓

Language that humans speak → stored as text or audio

Applications :

- sentiment analysis
  - given text / audio identify its sentiment
  - the text could be - positive, negative, neutral



- text classification

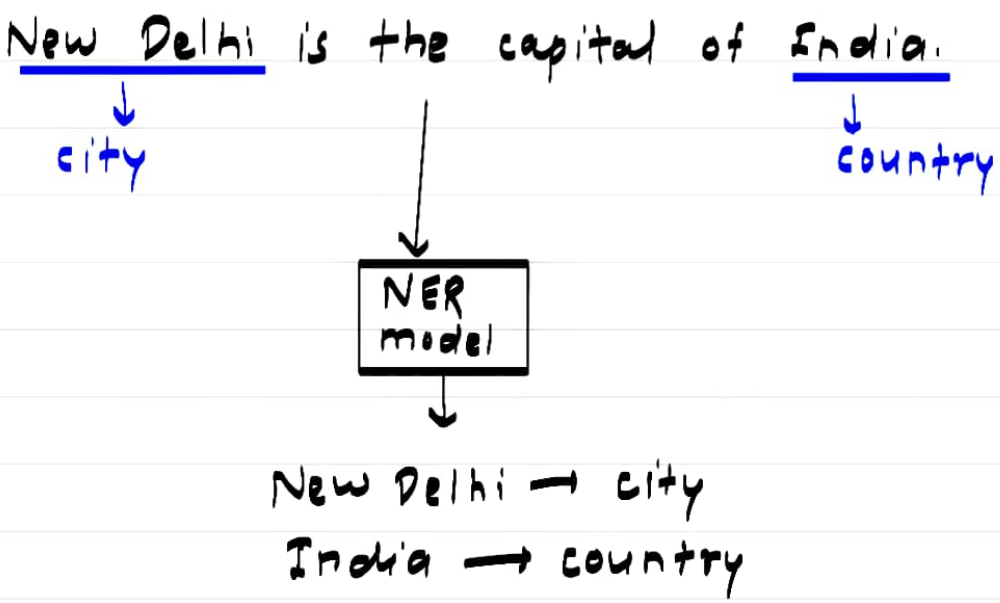
eg.

text	Topic
t_1	sports
t_2	weather
t_3	geopolitics
⋮	⋮
t_n	



* Token classification :

eg. named entity recognition



* Question Answering systems
  - chatgpt, gemini, deepseek, alexa, siri

  these are created using sequence to sequence models

* Text summarization
* Machine language translation



* Grammer correction
* Text to speech : convert text to audio
* Speech to text : convert audio to text

Lets understand NLP using Text classification

We started with using Bag of Words for text classification

Text	Labels
food is delicious and tasty	+ve
I liked the service	+ve
tables were not clean	-ve
parking was too small	-ve
restaurant is open	neutral

After all text preprocessing steps (tokenization, stop words removed, stemming/ lemmatization) we get vocab.

vocab = { food, delicious, taste, like, serve, table, clean, park, small, restaurant, open }

using the vocab we get numerical representation of the text

eg. I liked the tasty food at this restaurant

↓ text preprocessing

[like, taste, food, restaurant]

↓ BOW vocab

[1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0] vector form of text

vector form of text → Log regr, DTree, ANN, RF, etc → +ve / -ve / neutral